CSA06507-Generative AI

Lab Experiment's

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Unit 3: Lab Programs for Prompting

Experiment 3: Build a Vector Database Using FAISS and Perform Similarity-Based Document Retrieval

Aim:

To build a vector database using FAISS (Facebook AI Similarity Search), convert text documents into numerical embeddings, and perform similarity-based document retrieval using a user query.

Algorithm

1. Start the program.
2. Install and import the required Python libraries.
3. Create a collection of sample text documents.
4. Load the Sentence Transformer model.
5. Convert the documents into numerical embeddings.
6. Create a FAISS vector database.
7. Store the document embeddings in the FAISS index.
8. Accept a search query from the user.
9. Convert the query into an embedding.
10. Search for similar documents using FAISS.
11. Display the most relevant documents and similarity scores.
12. Stop the program.

Python Program

Step 1: Install libraries

Run this command in Google Colab.

Python

Run

!pip install faiss-cpu sentence-transformers

Step 2: Python Code

Copy and run this code in another Colab cell.

Python

Run

```
# Vector Database Using FAISS

import faiss
import numpy as np
from sentence_transformers import SentenceTransformer

# Step 1: Load the model
model = SentenceTransformer('all-MiniLM-L6-v2')

# Step 2: Create sample documents
documents = [
    "Artificial Intelligence helps machines learn.",
    "Python is a programming language.",
    "Machine Learning is a part of AI.",
    "Deep Learning uses neural networks.",
    "Computer networks connect computers."
]

# Step 3: Generate document embeddings
embeddings = model.encode(documents)

# Step 4: Convert embeddings into NumPy array
embeddings = np.array(embeddings, dtype='float32')

# Step 5: Normalize vectors
faiss.normalize_L2(embeddings)

# Step 6: Create FAISS index
dimension = embeddings.shape[1]

index = faiss.IndexFlatIP(dimension)
```

Step 7: Add documents to database

index.add(embeddings)

print("FAISS Vector Database Created Successfully!")

Input

Example input:

Plain text

Enter your search query: Artificial Intelligence and Machine Learning

Output

The output will be similar to the following. Exact similarity scores may vary.

Plain text

FAISS Vector Database Created Successfully!

Enter your search query:

Artificial Intelligence and Machine Learning

TOP 3 SIMILAR DOCUMENTS:

Document: Machine Learning is a part of AI.

Similarity Score: 0.xxxx

Document: Artificial Intelligence helps machines learn.

Similarity Score: 0.xxxx

Document: Deep Learning uses neural networks.

Similarity Score: 0.xxxx

Document Retrieval Completed!

Explanation

1. Vector Database:

A vector database stores numerical representations of text, images, and other data for similarity-based searching.

2. FAISS:

FAISS is a library used to store, organize, and search numerical vectors efficiently.

3. Embeddings:

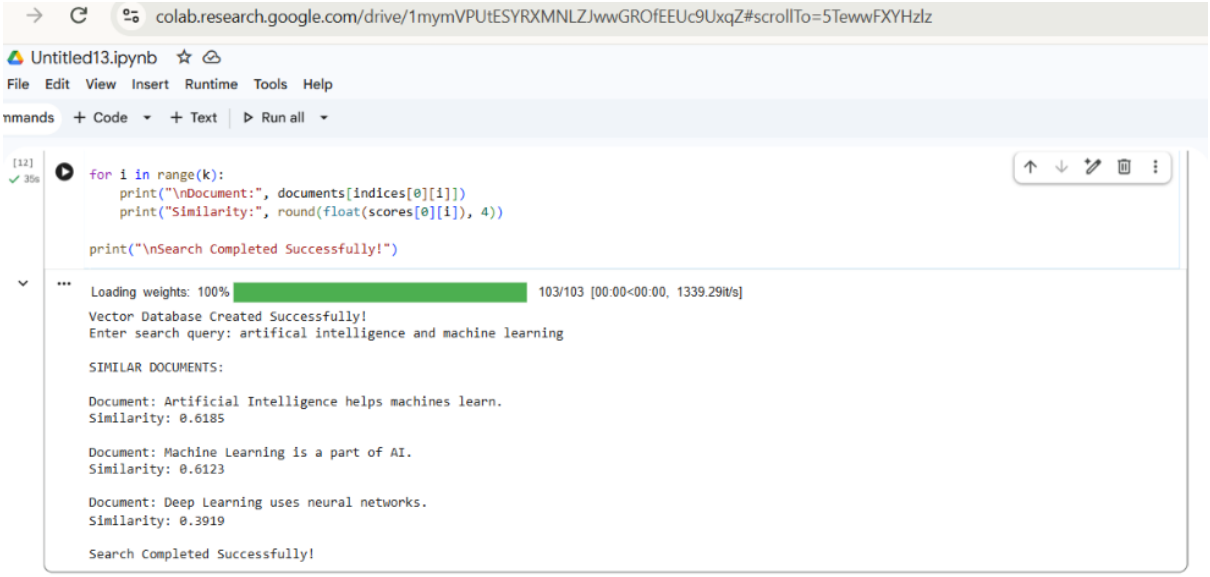
Embeddings convert text documents into numerical vectors that represent their meaning.

4. Similarity Search:

FAISS compares the query vector with stored document vectors and retrieves the most similar documents.

5. IndexFlatIP:

It performs inner-product similarity search. Since the document and query vectors are normalized, the inner product represents cosine similarity.



Result

Thus, a vector database was successfully created using FAISS in Google Colab. Text documents were converted into numerical embeddings and stored in the FAISS index. Similarity-based document retrieval was performed successfully, and the top three relevant documents were displayed based on the user's search query.